SECOND PUBLIC EXAMINATION

Honour School of Mathematics Part C: Paper SC7
Honour School of Mathematics and Statistics Part C: Paper SC7
Honour School of Mathematics and Computer Science Part C: Paper SC7
Master of Science in Mathematical Sciences: Paper SC7

BAYES METHODS

TRINITY TERM 2021

Thursday 3 June

Opening time: 09:30 (BST)

Mode of completion: Handwritten

You have 1 hour 45 minutes writing time to complete the paper and up to 30 minutes technical time to upload your answer file.

You may submit answers to as many questions as you wish but only the best two will count for the total mark. All questions are worth 25 marks.

You should ensure that you observe the following points:

1. Write with a black or blue pen OR with a stylus on tablet (colour set to black or blue).
2. On the first page, write
   - your candidate number,
   - the paper code,
   - the paper title,
   - and your course title (e.g. FHS Mathematics and Statistics Part C),

   but **do not** enter your name or college.
3. For each question you attempt,
   - start writing on a new sheet of paper,
   - indicate the question number clearly at the top of each sheet of paper,
   - and number each page.
4. Before scanning and submitting your work,
   - on the first page, in numerical order, write the question numbers attempted,
   - cross out all rough working and any working you do not want to be marked,
   - and orient all pages in the same way.
5. Submit all your answers to this paper as a **single PDF** document.

1. For $n \geq 1$ let $V_n = \{1, ..., n\}$ and let $\mathcal{E}_n = \{(i,j) : 1 \leq i < j \leq n\}$ be the set of all undirected edges (i,j). For $E \subseteq \mathcal{E}_n$ let (V, E) be a graph with vertex-set V and edge-set E.

For $i = 1, ..., n$ let $x_i \in [0,1]^2$ be n distinct points in the unit square and let $x = (x_1, ..., x_n)$. For $(i,j) \in E$ let $\ell_{(i,j)}$ be the edge-segment connecting x_i and x_j, so that

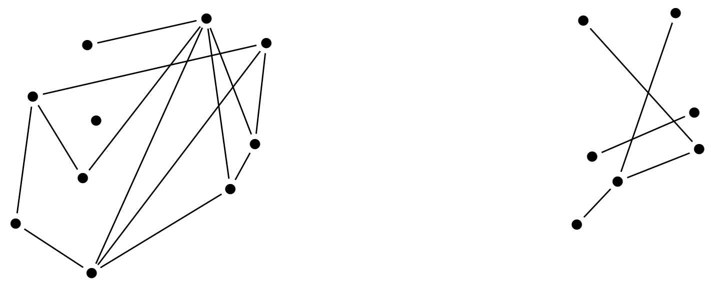
$$\ell_{(i,j)} = \{x_i + c(x_j - x_i) : 0 < c < 1\}.$$

For given $x, (V, E)$, let

$$L(x, V, E) = \{\ell_{(i,j)}\}_{(i,j) \in E}$$

be the set of line segments determined in this way. We call $Y = L(x, V, E)$ a "straight-line-graph". Let $\mathcal{G}_x = \cup_{E \subseteq \mathcal{E}_n} \{L(x, V, E)\}$ be the set of all straight-line graphs on x. Let $\mathbb{I}\{A\}$ be the indicator for A and let $\#Y$ equal the number of "edge-segment crossings",

$$\#Y = \frac{1}{2} \sum_{\ell \in Y} \sum_{\substack{\ell' \in Y \\ \ell' \neq \ell}} \mathbb{I}\{\ell \cap \ell' \neq \emptyset\}.$$



Above, two straight line graphs: (left) $n = 10$ and $\#Y = 6$; (right) $n = 7$ and $\#Y = 3$.

(a) [9 marks] Suppose $x = (x_1, ..., x_n)$ is fixed. Consider the observation model

$$p(y|\theta, x) = \exp(-\theta \#y)/Z(\theta, x), \quad y \in \mathcal{G}_x,$$

for $\theta > 0$ a parameter penalising crossings and $Z(\theta, x)$ a normalising constant. In the following we drop "x" dependence from the notation when it is fixed.

Let $Y \sim p(\cdot|\theta)$, $Y \in \mathcal{G}_x$ and suppose we observe $Y = y_{obs}$. A prior $\pi(\theta)$ for θ is given.

(i) Write down the posterior for $\theta | y_{obs}$ and explain why it is doubly intractable.

(ii) Consider Approximate Bayesian Computation (ABC). Specify and briefly justify a summary statistic $S(y)$, $y \in \mathcal{G}_x$ and write down the ABC posterior $\pi_{ABC}(\theta | y_{obs})$ in terms of the model elements $\pi(\theta), p(y|\theta)$ and $S(y)$ and a threshold $\delta > 0$.

(iii) Suppose an algorithm simulating $Y \sim p(\cdot|\theta)$ is available. Give an ABC-rejection algorithm targeting $\pi_{ABC}(\theta | y_{obs})$.

(b) [9 marks] Give an MCMC algorithm targeting $Y \sim p(\cdot|\theta)$ and show it is irreducible. [*Hint: choose an edge at random from $\mathcal{E}_n$; add it if absent and otherwise remove it.*]

(c) [7 marks] Let $\theta > 0$ and $q \in (0,1)$ be given. For $n \in \{0, 1, 2, ...\}$, $x \in \mathbb{R}^{2n}$, $y \in \mathcal{G}_x$, and $f(x)$ a given function of x, consider the probability distribution

$$p(n, x, y | q, \theta) \ \propto \ f(x) \ \exp(-\theta \#y) \ n! \ q \, (1-q)^{n-1}.$$

Here $f(x)$ is a positive function ensuring $p(n, x, y | q, \theta)$ is finitely normalisable.

Give the updates adding and deleting points from x in reversible jump MCMC targeting $p(n, x, y | q, \theta)$. [*Hint: one point in the figure above would be particularly easy to delete.*]

2. When farming began, it is thought smaller settlements farmed more intensively, and in particular fertilized their crops more heavily. Archaeologists interested in testing this hypothesis characterised fertilizer levels as "low" (0), "medium" (1) or "high" (2). Fertilizer level changes the nitrogen isotope ratio in seeds from plants. The archeologists gathered n seeds from excavations at settlement sites with different settlement area sizes.

For $i = 1, ..., n$ let $x_i > 0$ give the area of the settlement where the i'th seed was excavated and let $y_i \in \mathbb{R}$ give the nitrogen isotope ratio measurement. This is normal with a mean $\mu_{f_i} = E(y_i),\ f_i \in \{0, 1, 2\}$ and $-\infty < \mu_0 < \mu_1 < \mu_2 < \infty$ and standard deviation $\sigma > 0$.

A model for the unknown fertiliser level $f_i \in \{0, 1, 2\}$ the plants received is given in terms of a latent variable $z_i \in \mathbb{R},\ i = 1, ..., n$ and two unknown thresholds, $-\infty < c_1 < c_2 < \infty$. The observation model is as follows

$$\begin{aligned}
&z_i \sim N(\beta x_i, 1)\\
&f_i = \mathbb{I}_{z_i > c_1} + \mathbb{I}_{z_i > c_2}\\
&y_i \sim N(\mu_{f_i}, \sigma^2), \text{ all jointly independent for } i = 1, ..., n.
\end{aligned}$$

The archaeologists have some modern isotope ratio data $v_j,\ j = 1, ..., m$, in which the manure levels $g_j \in \{0, 1, 2\},\ j = 1, ..., m$ are known but there are no settlement-area values. The observation model for the modern data is

$$v_j \sim N(\mu_{g_j}, \sigma^2), \text{ jointly independent for } j = 1, ..., m.$$

(a) [10 marks] Let $y = (y_1, ..., y_n),\ z = (z_1, ..., z_n),\ v = (v_1, ..., v_m)$ and $g = (g_1, ..., g_m)$. Let $c = (c_1, c_2)$ and $\mu = (\mu_0, \mu_1, \mu_2)$.

- (i) What factors would you consider in choosing a prior for β and how would you check it represented available prior knowledge?
- (ii) Suppose a joint prior $\pi(\beta, c, \mu, \sigma)$ is given. Write down the posterior $\pi(\beta, c, \mu, \sigma, z|y, v, g)$ in terms of the model elements, up to a constant factor.
- (iii) How would you test the hypothesis raised by the archaeologists?

(b) [15 marks] Suppose we take a simple approach, estimating $\widehat{\mu}_0, \widehat{\mu}_1, \widehat{\mu}_2$ and $\widehat{\sigma}$ with their MLE's from the modern data v, g and then setting

$$\widehat{f}_i = \arg\max_{k=0,1,2} N(y_i; \widehat{\mu}_k, \widehat{\sigma}^2)$$

for $i = 1, ..., n$. Let $\widehat{f} = (\widehat{f}_1, ..., \widehat{f}_n)$. These $\widehat{f}$-estimates are now taken as fixed data.

- (i) Consider estimating β with an estimate of the posterior mean $E_{z,c,\beta|\widehat{f}}(\beta)$. Identify a weakness of this approach. Give a setting where it would nevertheless lead to good approximate inference for β.
- (ii) Write down the posterior $\pi(z, c, \beta|\widehat{f})$ in terms of the model elements and a prior $\pi(c, \beta)$ for c and β, up to a constant factor.
- (iii) Calculate the marginal posterior $\pi(\beta, c|\widehat{f})$ giving your answer as an explicit function of c, β and $\widehat{f}$ and $x_1, ..., x_n$, and involving the CDF Φ of the standard normal.
- (iv) Given a choice between carrying out the inference using MCMC targeting $\pi(z, c, \beta|\widehat{f})$ or MCMC targeting $\pi(c, \beta|\widehat{f})$, what factors would you consider in making a choice?

3. (a) [11 marks] For $p \geq 1$, real $\alpha > 0$ and H a continuous distribution over $\Omega = \mathbb{R}^p$, a random distribution $G \sim \Pi(\alpha, H)$ on Ω has a Dirichlet Process distribution with parameter α and base distribution H. Given G and $n \geq 1$, parameters $\theta_i \sim G$, $i = 1, 2, ..., n$ are jointly independent and distributed according to G. Let $\theta = (\theta_1, ..., \theta_n)$.

(i) The Multinomial Dirichlet Process and the Dirichlet process are related. State the relation and what insight it gives us as to the character of the Dirichlet process.

(ii) State briefly why the Dirichlet process may be useful for clustering.

(iii) Suppose $n \geq 10$. Explain how to simulate θ_9 and θ_{10} without simulation of $G \sim \Pi(\alpha, H)$.

(iv) Calculate the probability that $\theta_7 = \theta_8$ and (in the same realisation) $\theta_9 = \theta_{10}$.

(b) [14 marks] Suppose now we have two independent Dirichlet processes $G_1 \sim \Pi(\alpha_1, H_1)$ and $G_2 \sim \Pi(\alpha_2, H_2)$ with $\alpha_1, \alpha_2 > 0$ and H_1 and H_2 continuous distributions over $\mathbb{R}$ with densities h_1 and h_2 respectively.

Let $\theta_{1,j} \sim G_1$ and $\theta_{2,j} \sim G_2$ all jointly independent given G_1 and G_2 for $j = 1, ..., n$. For $i = 1, 2$, let $\theta_i = (\theta_{i,1}, ..., \theta_{i,n})$. For $j = 1, ..., n$ let $x_j \in \mathbb{R}$ be a scalar covariate. Consider the observation model

$$y_j = \theta_{1,j} + \theta_{2,j} x_j + \epsilon_j, \quad \epsilon_j \sim N(0, \sigma^2),$$

with ϵ_j jointly independent for $j = 1, ..., n$. A prior $\pi(\sigma)$ for σ is given.

(i) Explain qualitatively, perhaps with the aid of a graph plotting y against x, what sort of clustering is modelled here.

(ii) For $i = 1, 2$ let $\theta_i^* = (\theta_{i,1}^*, ..., \theta_{i,K_i}^*)$ give the K_i unique parameter values in $\theta_i = (\theta_{i,1}, ..., \theta_{i,n})$. For $i = 1, 2$ let $S_i = (S_{i,1}, ..., S_{i,K_i})$ where for $k = 1, ..., K_i$,

$$S_{i,k} = \{j : \theta_{i,j} = \theta_{i,k}^*, j = 1, ..., n\}.$$

Give the posterior distribution $\pi(\theta_1^*, S_1, \theta_2^*, S_2, \sigma | y)$ in terms of the model elements up to a constant factor. There is no need to write the prior distributions of S_1 and S_2 explicitly, but you should identify them with a known distribution.

(iii) Give the cluster-membership update in a Gibbs sampler targeting $\pi(\theta_1^*, S_1, \theta_2^*, S_2, \sigma | y)$. It is sufficient to give the update for the cluster assignment of j in S_1 for one $j \in \{1, ..., n\}$.